Anay Baid

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Technical Skills

Languages & Frameworks: Python, R, Java, C, C++, JavaScript, TypeScript, SQL, Kotlin, C#, React.js, Node.js
Backend: Spring Boot, FastAPI, Flask, REST APIs, Kafka, PostgreSQL, Oracle, MongoDB, Redis
Cloud & Infra: AWS, Azure, S3, ECS/EKS, Docker, Kubernetes, Terraform, Linux, Git, Airflow, Playwright

Professional Work Experience

Software & Network Optimization Engineer — Purolator Inc *May 2026 - December 2026*

- Cut planning turnaround 10-20× and diagnostics 40% by owning backend development of **Python/PowerShell ETL pipelines**, **Airflow DAGs**, and **Flask REST APIs** processing **100K+ forecast records** with **SQL-backed state**, validation, and reconciliation.
- Containerized forecasting workloads with **Docker**, using **Amazon S3 + Parquet** for forecast artifacts and **Amazon SageMaker** for managed inference; deployed via **ECR/ECS/Fargate**.
- Provisioned the **Airflow → S3 → SageMaker → ECS/Fargate** architecture with **Terraform**, IAM-scoped workloads, private networking, **Secrets Manager**, and **KMS**.
- Developed **Python constraint-aware allocation algorithms** preserving network volume targets, exposing capacity bottlenecks and supporting **10-15% projected throughput gains**.

Software Engineer — Iotum Global Holdings Inc *May 2025 - August 2025*

- Engineered full-stack **React reference applications** to demonstrate complex **SDK integrations**, increasing external developer adoption & reducing client implementation time by **30%**.
- Architected scalable test automation frameworks using **JavaScript**, **Selenium**, and **Mocha**, increasing code coverage by **85%** & enabling faster **CI/CD release cycles**.
- Resolved critical **user-facing latency issues**, improving application stability under high concurrency for **10,000+** users.

DevOps Engineer — Nationwide Appraisals *September 2024 - December 2024*

- Modernized legacy infrastructure by **containerizing services with Docker** and optimizing **TCP/IP and Layer 3 networking**, improving deployment consistency and application performance across enterprise environments.
- Automated secure cloud and mobile infrastructure workflows across **AWS, Azure, and Intune MDM**, using **SQL validation**, encrypted **SSH tunneling**, Java/Android tooling, and device provisioning automation to reduce manual setup overhead by **40%**.

Data Engineer — Cineplex Entertainment LP *January 2024 - April 2024*

- Engineered real-time **ETL pipelines** for the **Cineplex Homepage Dashboard**, ensuring high-availability box office reporting via automated data validation scripts.
- Optimized **Machine Learning algorithms** for show-time scheduling, improving prediction accuracy by **15%** via rigorous model evaluation & input data cleansing.
- Engineered automated **Python** data-validation pipelines with **SQL schema comparison** and **JUnit/Selenium** test automation, reconciling **10TB+ of Azure & On-Prem data**, detecting and resolving **200+ data anomalies**, and maintaining **100% data integrity**.

Hackathons & Project Experience

InsightVault (Financial Fraud Detector) | *Java, Spring Boot, React, TypeScript, AWS, Terraform* *August 2026*

- Engineered a fraud-alert triage API in **Java/Spring Boot**, blending an auditable rule engine with a trained **logistic regression** model to hit **92.9% precision** / **94.1% recall** on a **35K+**-transaction time-based holdout.
- Built a **Python** (**pandas, scikit-learn**) training pipeline exporting model weights to a dependency-free **Java** inference layer with **1e-6** train/serve tolerance, via a **React/TypeScript** dashboard with **Vitest/Playwright** tests, deployed on **AWS ECS Fargate, RDS, and CloudFront** via **Terraform**.

StarRez Enterprise Automation Suite | *JavaScript, DOM Injection, Workflow Automation* *February 2026*

- Architected a production-grade automation extension for **100+ Front Desk staff**, reducing manual logging by **80%** via automated DOM data extraction & processing pipelines.
- Designed a scalable, modular workflow engine in **JavaScript**, enabling campus-wide deployment approved by **University IST Management**; supports high-volume student record processing with zero manual intervention.

CI/CD Build Automation Tool | *Python, GitHub Actions API, SMTP, REST* *UOttaHack 5 (March 2023)*

- Won **Best Use of GitHub** by engineering a **Python** CI/CD orchestration tool that used the **GitHub API** to trigger targeted builds across environments, with automated **environment validation** and secure **SMTP notifications**.

Education

University of Waterloo — Honours Computer Science *Exp. April 2027*

- Achieved **100%** in **CS 445 (Software Requirements Specification) & CS 446 (Software Design and Architectures)**; recognized by the Director of Software Engineering.